Multimedia Tools and Applications (2025) 84:4163–4201
https://doi.org/10.1007/s11042-024-18801-z

Image splicing forgery detection: A review

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Received: 25 July 2023 / Revised: 5 October 2023 / Accepted: 24 February 2024 /
Published online: 16 March 2024
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Abstract
Image splicing forgery is a prevalent form of digital image manipulation where various portions from one or multiple images are combined to create a deceptive image that appears genuine. Detecting image splicing forgery is crucial for verifying the authenticity of an image. Image splicing forgery detection has grown significantly in recent years, with numerous detection approaches proposed in the literature. This paper presents a comprehensive survey and classification of existing image splicing forgery detection approaches, focusing on 2014 to 2023. This study reviews 88 research papers on splicing in the context of image forgery detection. A generalized structure is introduced, outlining the typical stages involved in the detection process. The paper thoroughly reviews the literature, providing an overview of both hand-crafted and advanced detection approaches researchers propose. Benchmark datasets are identified, including their limitations. The objective is to provide a clear and comprehensive understanding of image splicing forgery detection for researchers and practitioners interested in this area. This survey is a valuable resource, offering insights into the field's current state and highlighting areas for future research and development.

Keywords Image forgery detection · Image forensic · Splicing · Fake news · Deep learning · CNN

1 Introduction

The internet and social media are full of images. Millions of images transfer on social media from Facebook, and Twitter to Instagram, some of which may spread fake news. Image manipulations influence personal interest and design with a malicious intention to spread fake news in society. This false information creates distress and influences people's minds during important events like an election. Let's understand the impact of spliced images through an actual incident. In 2011, Indian Prime Minister Dr. Manmohan Singh

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met with US President Mr. Barack Obama at an ASEAN summit. But then, years later, a spliced image as shown in Fig. 2(g), was circulated on Twitter where Obama was shaking hands with President of Iran Hassan Rouhani [1]. This spliced image was circulated with malicious intentions on the internet years later, during the US election in 2020. Therefore, it is crucial to identify spliced images and stop their online circulation.

However, various methods exist to manipulate an image. Some of these methods are image retouching, image compression, copy-move, splicing, removing watermarks or digital signatures, etc. Among these techniques, image splicing stands out as particularly concerning due to its potential to completely alter the information or content within the image. Therefore, image forgery detection is always a potential topic among scholars, especially copy-move and splicing. Splicing is an image forgery method where some parts from one or many images combine to create a new image.

Some current surveys and review studies on image forgery and detection are included below where it is visible that most of the time, research studies cover splicing as a subpart of their work [2, 3]. The detailed classification of image forgery schemes is presented in [4] and focuses on preprocessing steps, camera irregularities, optical and sensor operation, and other statistical features. In another review, M. A. Qureshi and M. Deriche [5] discuss several pixel-based detection approaches for splicing, copy-move, image sampling, and retouching. The paper also includes several post-processing detection techniques such as median filtering, sharpening, contrast improvement, and noise detection techniques. The other studies [6–9] also compare various existing approaches for image forgery detection. In the paper [7], the authors extensively discuss four primary categories of tampering detection techniques: image splicing, copy-move forgery, image resampling, and image retouching detection. Furthermore, passive image forgery detection is categorized into two sections; source camera identification-based forgery and tempering-based forgery [8]. An Interpol study on forensic images and video analysis in 2020 [10] mainly covers the literature on imaging and video forgeries from 2016–19. The authors in [11] review different detection algorithms for splicing and copy-move forgery including intensity-based, feature reduction-based, and key-feature-based algorithms. The study also provides future directions on Convolutional Neural Networks (CNN) and highlights the limitations of solution approaches.

Deepfakes are spoofing methods designed using artificial intelligence. These methods gather data on physical movements, facial characteristics, and even voice patterns. This data is then processed using an AI-encoded Generative Adversarial Network (GAN) to generate fake yet remarkably real-like audiovisual content. Generally, deepfakes work more on face substitutions than body gestures or postures. The extensive and dedicated analyses of deep learning approaches for forgery detection are also presented in [12] which includes networks such as CNN, Visual Geometry Group (VGG), Residual Networks (ResNet), GAN, U-shaped encoder-decoder network (Unet), and other networks (explicitly discussed in Section 3.2). The survey [13] provides a wide-ranging classification of forgery, a comparison of various detection approaches for image forgery, and also offers a comprehensive overview and analysis of deep learning-based forgery detection methods.

But as the findings, above mentioned review papers include splicing as a sub-topic under image forgery detection. Some of these review papers are also listed in Table 1. Therefore, this work focuses and covers the recent development only on splicing detection on images and presents a hierarchical categorization of existing solution approaches for splicing detection.

The keen interest of scholars and young researchers is visible from the number of papers published in the past ten years. In this study, almost all existing detection approaches are

Table 1 Existing review papers on image forgery detection

References	Year	Image Forgery Detection
Zanardelli M et al. [12]	2022	Deep learning approaches Image forgery detection
Meena K. B. et al. [7]	2019	Image Forgery Detection: Survey and Future Directions
Panda S. et al. [6]	2018	Passive Techniques of Digital Image Forgery Detection
Asghar K. et al. [11]	2016	Copy-move and splicing image forgery detection and localization techniques
Qureshi, M. A. et al. [5]	2015	A bibliography of pixel-based blind image forgery detection techniques
Birajdar G. K. et al. [4]	2013	Digital image forgery detection using passive techniques: A survey

incorporated and presented in the last ten years on splicing. Figure 1 shows the total number of papers on splicing detection published in Springer, IEEE, and Elsevier in the past 10 years. It is important to mention that for the search query and for the rest of the literature, the terms forgery, tempering, manipulation, and fake resemble the same meaning. The graph in Fig. 1 shows two important aspects of this research so far. First, this topic is trending among scholars, and second, there is massive scope for improvement in this field. As per the stats, there is a big difference between the number of papers published in journals and conferences.

1.1 Contribution and organization of the paper

A number of existing review papers are discussed in the above section which clearly states that there is a potential scope to study splicing and its detection approaches separately. Therefore, this study presents a comprehensive review of existing research on splicing

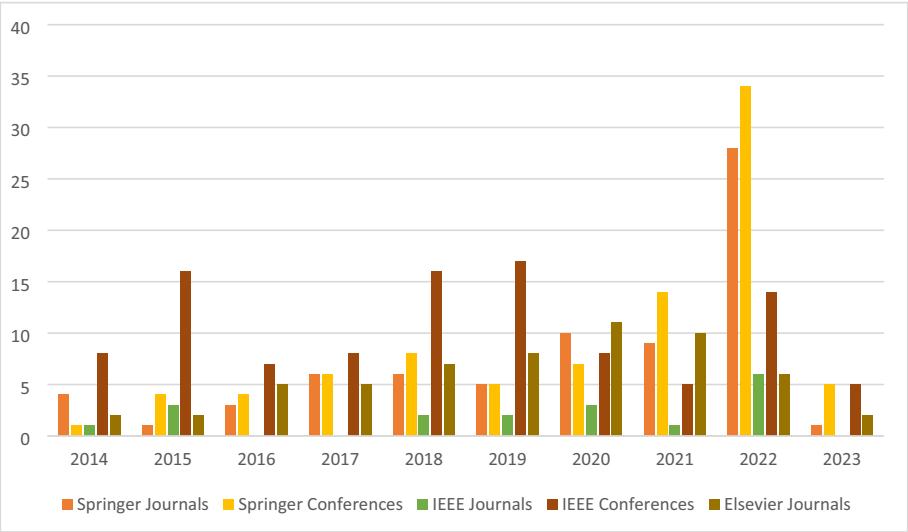

Fig. 1 Number of publications over the past 10 years. (The results are collected using the query 'Image splicing detection' from Springer, IEEE, and Elsevier websites)

detection from 2014 to 2023. It analyzes the current approaches and recent advancements, aiming to address the following key issues:

- The study presents the first-ever review on Image Splicing Forgery Detection (as per author's knowledge).
- The study comprehensively reviews 88 research papers to identify the traditional and state-of-the-art approaches in splicing detection.
- The work also tabulates all the existing datasets with their characteristics and limitations.

A comprehensive overview of the existing literature, and insights into the challenges faced in implementing a generalized structure for splicing detection are presented. This study is a valuable resource for researchers and practitioners interested in this field, facilitating further advancements in splicing detection techniques. Overall, this review tries to address some important questions pertaining to image splicing forgery detection are given below:

- RQ1: Which hand-crafted approach is the most effective for localization of splicing?
- RQ2: Which Deep-learning approach is the most effective for localization of splicing?
- RQ3: Which classifier is the most effective for the classification of spliced images?
- RQ4: Which dataset is suitable for splicing?
- RQ5: What parameter should be chosen for evaluation metrics?

The rest of the paper is structured in the following manner. Section 2 introduces image forgery and its types with relevant examples. Section 3 explains the image splicing detection process and categorizes the splicing detection techniques into hand-crafted and advanced deep-learning techniques. The literature combines several feature-based forgery detection approaches. Section 4 provides the details regarding the dataset available for splicing detection. Section 5 lists the broad classification of current solution approaches in block-based, key feature-based, and learning-based approaches with their limitations on advanced threats. Section 6 concludes the study, and the last Section 7, focuses on future work and emerging fore-coming challenges.

2 Image forgery: Types and detection techniques

In this section, various forms of image forgery are examined through relevant examples. However, the primary emphasis is directed toward image splicing only, understanding how it is different from other forms of forgery, and delving into the basic process for its localization within an image. Image forgery refers to the act of manipulating or altering digital images with the intent to deceive or mislead viewers. It involves making unauthorized changes to an image to create a false representation of reality. Image forgery can take various forms, such as combining elements from different images to create a new, enhancing image called retouching, cloning or duplicating objects within the same image, deleting objects or people from an image, and watermarking removal from the original images. Image forgery can be broadly categorized into active and passive image forgery methods [2]. Various image forgery methods are provided in Fig. 2, where many forgery methods are applied to the same image for better understanding.

(a) Original Image (b) Watermark (c) Retouching

(d) Resampling (e) Compression (f) Copy-move

(g) Spliced image (third one) made with two original images (first and second)

Fig. 2 Different types of image forgery methods performed on the original Image 2(a) [1], (**a**) Original Image, (**b**) Watermark, (**c**) Retouching, (**d**) Resampling, (**e**) Compression, (**f**) Copy-move, (**g**) Spliced image (third one) made with two original images (first and second)

2.1 Active forgery methods

It involves the insertion of a unique key into the original image before its online circulation. This pre-inserted key or code serves as a means to verify the image's authenticity. However, this technique is only effective for images circulated for the first time online. The active forgeries can further be divided into two methods given as follows.

a) **Digital Watermarking**: Watermarking involves the insertion of a marker within digital media, including audio, video, or images as shown in Fig. 2(b). Its purpose is to ascertain the copyright ownership of the content. Digital watermarking is used to track instances of copyright infringement on the internet and to verify the authenticity of currency notes within the banking system. These watermarks can be visible or concealed markers.
b) **Digital Signature:** A digital signature is not just a digitized certificate. It employs mathematical techniques to confirm the genuineness and integrity of a digital document. Unlike a conventional signature that comprises the user's name, a digital signature consists of two distinct keys (private key for sender and public key for receiver) or sequences

of characters. It utilizes cryptographic methods to secure the content of a message or document.

2.2 Passive forgery methods

It refers to techniques that do not require prior information or a specific key for verification. Passive forgery is crucial as only a small percentage of images, typically ranging from 2 to 5%, are actively secured with such keys. Moreover, with the advancements in digital image forgery, including deepfake technology, identifying artifacts of image manipulation has become increasingly challenging. Therefore, passive image forgery methods are crucial in detecting and addressing image manipulations, even when active measures are not implemented.

a) **Image Retouching:** It refers to the process of enhancing digital images to improve their quality and correct imperfections to make them look more aesthetic. This technique is commonly used in photography, advertising, fashion, and on social media platforms such as Facebook or Instagram where image appearance is a bit critical. However, retouching is generally regarded as a non-serious form of image manipulation, because it involves, color correction, restoration, background enhancement, and skin smoothening. The image in Fig. 2(c) is a retouched image where the stars (in the original image) are removed from the background.
b) **Image Resampling:** Resampling involves altering the pixel dimensions of an image, either by adding or reducing the total number of pixels as shown in Fig. 2(d). This process is mainly applied for onscreen display purposes. It's crucial to note that resampling invariably results in a reduction of image quality. It's important to distinguish between resampling and resizing because resizing pertains more to adjusting the print size while resampling relates to changing the image's resolution.
c) **Image Compression:** Compression aims to reduce the file size of the image. Typically, image compression achieves this by eliminating certain data from the image as shown in Fig. 2(e). However, excessive compression can lead to a noticeable loss in image quality which often results in coarse pixels, a reduction in the number of colors, and a loss of fine details. In cases where a manipulated image undergoes multiple JPEG compressions, detecting the manipulations becomes significantly challenging due to the compound effect of compression artifacts [14].
d) **Copy-move:** Copy-move image forgery is a form of image manipulation where a portion of an image is duplicated and pasted within the same image. This manipulation typically leads to the creation of duplicate objects or elements within the image. However, it's important to note that the artifacts associated with the copied part of the image may differ from those present in the rest of the image [15, 16]. The image in Fig. 2(f) is a copy-move image where the stars (in the original image) are copied in the background.
e) **Splicing:** Splicing refers to the forgery in which parts of one or many images are copied and pasted to generate a whole new image. It is one of the most potential forgery methods that can change the information of the image. This technique often hides or removes particular objects or elements from an image while maintaining its overall appearance. Splicing aims to create an image that appears genuine but, in reality, is false. It includes cut-paste and blending techniques. Therefore, it is known as "cut-and-paste splicing," which involves copying a portion of one image and pasting it onto another image.

Figure 2 (g) is the perfect example of image splicing, where some parts of the first two original images [1, 17] are pasted together to create a new but fake spliced image.

2.3 Why image splicing is different and more challenging?

- In image retouching, resampling, compression, and watermarking manipulation are applied to the whole image, whereas in splicing and copy-move, the manipulations are oriented to the local region.
- Splicing involves combining elements from multiple images to create a new composition, while copy-move forgery involves duplicating and pasting regions within the same image.
- Splicing aims to create a fraudulent yet plausible scene representation, whereas copy-move forgery often seeks to duplicate or add redundant elements within the same image.

When an image is captured from a camera, it goes through many processes known as the image acquisition process as shown in Fig. 3. In this process, a light source illuminates the object to be captured. The illuminated object reflects light, containing visual information about the object. The lens allows the light to pass through the sensor. In digital cameras, a color filter array is positioned over the sensor for capturing RGB color information. The sensor then detects the pixel intensity of light, with color information usually inferred from neighboring pixels featuring different color filters. The raw sensor data undergoes demosaicing, where interpolation algorithms fill in missing color details to create a complete RGB image. Subsequently, color correction adjusts the colors in the image to ensure a true reproduction of the original scene. Finally, gamma correction is applied to fine-tune brightness levels, rendering the image visually consistent and more natural.

During this whole process, the image holds some intrinsic characteristics or artifacts as shown in Fig. 3, such as lens aberration produced due to minor defects in the lens, interpolation, sensor noise caused by manufacturing errors in sensors, post-processing artifacts produced due to corrections and lossy compression artifacts. These artifacts are consistent for a real image but for a manipulated image, these artifacts are found inconsistent. The image forgery can be detected using such inconsistencies.

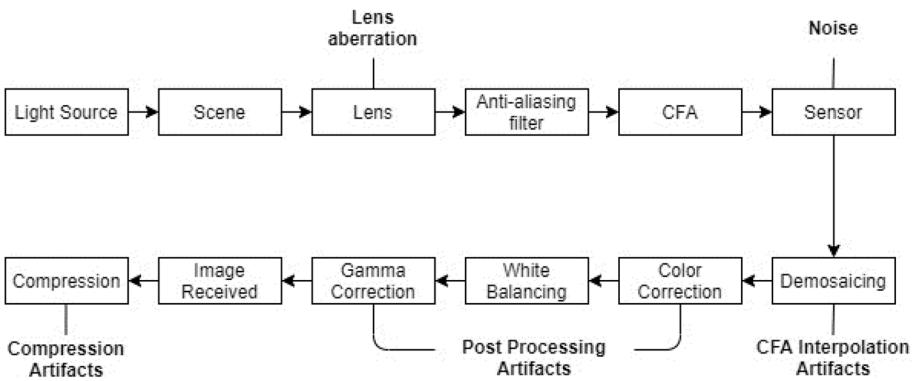

Fig. 3 Several steps of an image's life cycle and respective artifacts

3 Splicing detection approaches

This section includes several types of splicing detection techniques based on features. Splicing detection means localizing the splicing region of a manipulated image. This localization of forgery can be object-based, block-based, pixel-based, or image-based depending on the applied approach. Achieving this localization requires the identification of artifacts within the image induced by factors such as the camera, editing software, capturing lighting, and background details. Once the feature is selected, the subsequent steps involve choosing and implementing feature extraction and classification methods to detect the spliced area. Figure 4 shows the basic procedure of splicing detection with two fundamental steps; feature extraction and classification. It's worth mentioning that detection methods and algorithms may vary depending on the approach and research being conducted. Different feature-based techniques, such as key feature-based and block-based, may require slightly different steps or variations in the process. Selecting features and algorithms depends on the requirements, dataset, and available resources.

Here are the general steps involved in splicing detection:

1. **Preprocessing:** Perform the necessary steps to prepare the image for analysis. It may include converting the image to an appropriate color space, resizing, denoising, or enhancing image quality.
2. **Feature Extraction:** Extract relevant artifacts from the image that can help detect spliced regions. Commonly used features include PRNU noise patterns, chroma values, texture descriptors, block artifacts, noise statistics, and histogram properties.
3. **Region Segmentation:** Segment the image into smaller regions or blocks for localized analysis. This step helps in focusing on specific areas and identifying inconsistencies or anomalies.
4. **Individual Region Analysis:** Analyze each region using the extracted features. Apply appropriate algorithms or techniques based on the chosen detection methods. For example, compare PRNU patterns between regions, check for chroma inconsistencies, examine block artifacts, analyze noise characteristics, or evaluate histogram properties.
5. **Decision Fusion:** Combine the results from individual region analysis to decide the presence or absence of splicing in the image.
6. **Post-processing:** Apply post-processing techniques to refine the detection results and reduce false positives or negatives. It also includes additional filtering, smoothing, or validation steps to improve detection accuracy.
7. **Reporting and Visualization:** Present the detection results meaningfully, such as highlighting spliced regions in the image or generating a report indicating the presence and location of splicing.

Although the above-mentioned procedure keeps changing, as it depends on the feature extraction method, classification method, and on type of localization. In this paper, the splicing

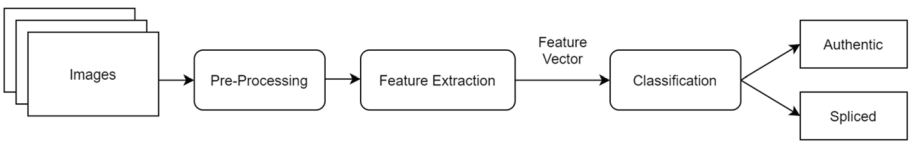

Fig. 4 Block diagram for detection of spliced image

detection approaches are categorized into two classes for simplification: handcrafted and deep-learning approaches. Traditional hand-crafted approaches use a hand-picked approach for feature extraction and classification. On the other hand, deep-learning approaches autonomously analyze images to identify spliced regions through self-learning, eliminating manual intervention. Some existing approaches in the literature hold great potential for enhancing the accuracy and efficiency of detecting splicing forgery.

3.1 Handcrafted splicing detection approaches

In this section, hand-crafted approaches are discussed and categorized into nine sub-sections depending on features; scenario-based approaches, format-based approaches, camera-based approaches, noise-based approaches, pixel-based approaches, geometric-based approaches, texture-based approaches, image quality metrics-based approaches, and light-based approaches.

Handcrafted approaches denote traditional forensic methods characterized by manual feature extraction and analysis. In these methodologies, forensic experts examined the particular features within an image to discern potential regions where splicing may have occurred. These features are noise, light, camera properties, image formats, scenarios, geometric patterns, textures, and pixels [11]. Splicing introduces inconsistencies in the feature patterns across different regions of an image. Some of the approaches analyze statistical properties of noise, to identify inconsistencies that may indicate splicing. Each camera sensor has a unique pattern noise that serves as a fingerprint to analyze this patterned noise to identify inconsistencies. Light-based techniques analyze illumination inconsistencies, such as differences in brightness, shadows, or reflections, to detect potential splicing. Scenario-based detection focuses on inconsistencies in shadow and blur, which is essential for scene understanding and object recognition tasks.

Inconsistent color properties, such as variations in color saturation, hue, or color distribution, indicate splicing. Unnatural or irregular edges and boundaries due to the blending of different image fragments can also be analyzed to detect splicing. When an image is compressed, artifacts that can reveal splicing may be introduced. These hand-crafted techniques require domain expertise and manually extracting relevant image features. A detailed categorization of current hand-crafted splicing detection techniques is shown in Fig. 5.

3.1.1 Scenario-based approaches

Scenario-based detection focuses on the objects in the image, their positions, and their textures. It helps understand an image's objects and lighting conditions. The two main components of scenario-based detection are as follows.

(a) Splicing detection using shadow: Shadow inconsistencies in a spliced image refer to discrepancies or abnormalities in the shadow patterns present within different regions of the manipulated image. These inconsistencies serve as potential indicators of image tampering. Shadow inconsistencies estimate the matte value, representing a spliced region's opacity or transparency. The matte value typically ranges between 0 and 1, representing the extent to which the spliced region blends with the surrounding area regarding radiance.

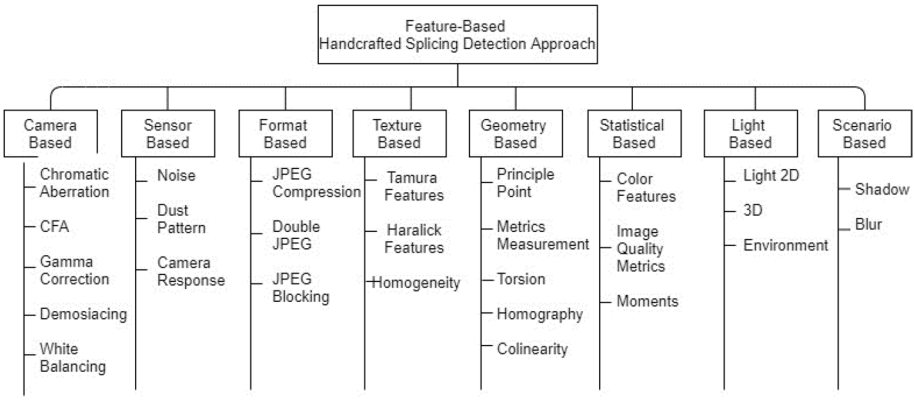

Fig. 5 Categorization of feature-based hand-crafted splicing detection techniques

B. Yang et al. [18] extracted the shadow boundaries using a Support Vector Machine (SVM) and calculated the shadow scale factor from the Growth Rate Penumbra Width (GRPW) as shown in Fig. 6. The model produced promising results (88.54% of TPR), but the major drawback is that the model only supports the image with good background light. **(b) splicing detection using Blur type inconsistencies:** Blur occurs for various reasons, such as camera shake, long exposure time, or defocus, which impact the overall quality of an image. The approach is to estimate the blur level of different regions within the image. Various blur estimation techniques can be applied to measure the amount of blur in specific areas. These techniques may involve analyzing the image's frequency content, edge sharpness, or gradient properties. Significant variations in the estimated blur levels between regions can indicate potential splicing. Blur is categorized into (a)

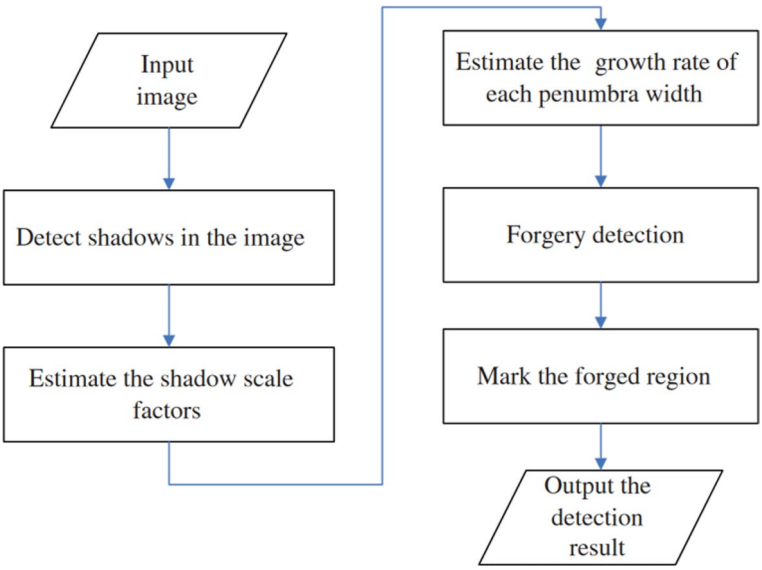

Fig. 6 The object-level localization of image splicing using shadow [18]

motion blur and (b) out-of-focus blur. Motion blur occurs due to the camera movement, while out-of-blur occurs due to the lens's inappropriate focal length. The advantages of blur detection over other detection techniques are as follows. (a) It is robust to resizing, and (b) imaging device information is not required. K. Bahrami et al. [19] proposed a framework for identifying blur inconsistencies by applying Bayesian-based Maximum A Posteriori (MAP) for local blur-type feature extraction followed by a binary classifier. M. P. Rao et al. [20] use the Fourier-Mellin Transform (FMT) for image registration and to calculate the orientation of each blur kernel. Slow shutter speeds of a camera cause motion blur, calculated by averaging the intensity of pixels over the camera's shutter time interval. Motion blur [21] inconsistencies locate discontinuity in an image.

3.1.2 Format-based approaches

This section includes format-based splicing detection which focuses more on analyzing the image file's metadata, such as the file format, compression method, and color space, etc. The JPEG is one of the most commonly used image compression formats used by cameras to reduce the storage space. These JPEG or double JPEG compression can be used to identify the splicing [22].

A basic example of a JPEG spliced image is shown in Fig. 7. The JPEG compression occurs in three stages: sampling, quantization, and entropy coding [24, 25]. In the sampling stage, the image is divided into blocks and then every block is transformed into color space. The quantization is applied to each block to generate a quantization matrix. Finally, entropy coding is applied to the quantization to produce the compressed image. It is important to note that some information is lost during the compression process, affecting the image's quality and suitability for spliced image detection.

For the detection of such spliced images, Zhu N. et al. [26] proposed a framework using a noise-free DCT model with a histogram as shown in Fig. 8. The block-level localization is used to highlight the forged region. The input image undergoes a transformation into the YCbCr color space. Subsequently, the Y map is partitioned into non-overlapping blocks to mitigate quantization noise. The localization is illustrated through a detailed block-wise posterior probability map.

(a) source image 1, actual size 800×1163 (b) source image 2, actual size 900×600 (c) tampered image, actual size 605 × 908

Fig. 7 An example of JPEG image splicing with resizing [23]

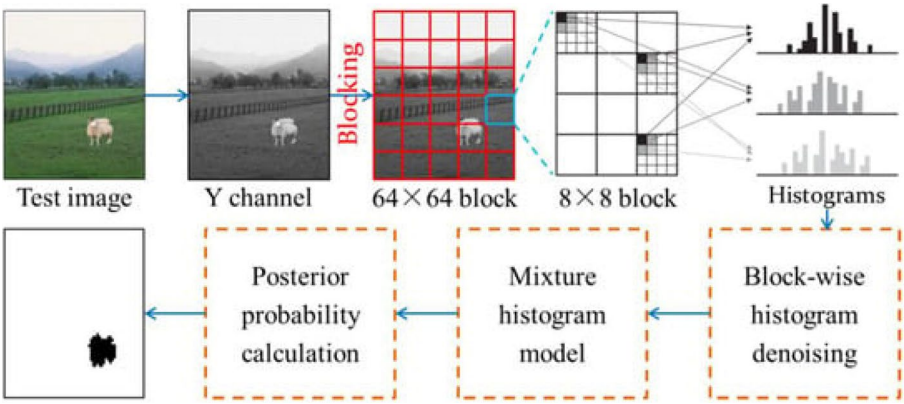

Fig. 8 A framework for JPEG image splicing detection [26]

3.1.3 Camera-based approaches

This section includes approaches based on artifacts introduced by cameras to localize splicing in an image. Understanding the properties of camera is essential for this approach. Knowing the device's characteristics makes it possible to determine the best way to process the image and extract relevant information. Camera-based detection techniques emphasize color correlation, camera response, white balancing, filtering array, and sensor noise for feature extraction.

Color Filter Array (CFA) [27] is a mosaic of tiny filters placed on top of image sensors in digital cameras and other imaging devices. The filters allow only specific wavelengths of light to pass through to the sensor, which can then detect the intensity and color of the light to create a digital image. Figure 9 shows a typical CFA-based splicing detection approach where all color channels (RGB) are used to estimate the kernel. When light passes through the CFA and reaches the sensor, the sensor records the intensity of each color in each pixel. After capturing the image, the software algorithms interpolate the missing colors to create a full-color image. Different cameras and imaging devices use different CFAs, and the CFA quality can significantly impact the final image quality.

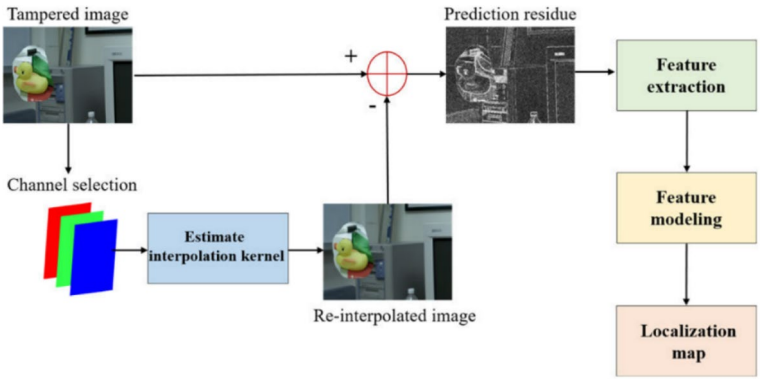

Fig. 9 CFA-based forgery localization suggested by C. W. Park et al. [28]

C. W. Park et al. [28] apply Singular Value Decomposition (SVD) to detect the image forgery. This approach aims to reconstruct the original image from the singular values of the image and compare the reconstructed image with the original image. For image-level classification between forged and real images, W. Wang et al. [29] use the color Croma (YC_bC_r) for an image with the help of a Gray-Level Co-occurrence Matrix (GLCM) for feature extraction and SVM.

3.1.4 Noise-based approaches

This section discusses methodologies that utilize inconsistencies in noise patterns to localize spliced regions. These approaches generally target pixels with noise characteristics. Noise-based detection [30] techniques calculate different types of noise, such as Fixed Pattern Noise (FPN), Poisson, and white Gaussian noise, to determine the forgery. Poisson noise is a type of noise that is inherent in image sensors and is proportional to the intensity of the light in the scene. On the other hand, White Gaussian noise [31] is a type of noise that is added to the image to conceal forgery. However, noise-based detection techniques are ineffective if the tempered image is consistent with the uniform noise pattern. FPN generates only where the sensor array is not exposed to light, which may be detected by calculating Photo Response Non-Uniformity (PRNU).

For noise-based detection, the initial aim is to separate noise signals to calculate noise function. B. Liu and C. M. Pun [32] propose an adaptive SVD to detect manipulation. They calculate a local multi-level noise map and a vicinity noise descriptor to identify the spliced region in the image. The noise level is higher in the spliced region than in the rest. However, the results suffer from low accuracy. In a different approach, the authors from [31] compute discrepancies in noise patterns across multiple scales for detecting splicing forgery using pixel inconsistencies. Simple Linear Iterative Clustering (SLIC) is used to segment the image into multiple scales, and Optimal Parameter Combination Searching (OPCS) is used to identify the critical features. The scheme is robust for multi-object splicing as well.

Image splicing in raw images can be localized by highlighting local noise inconsistencies. Based on this, T. Julliand et al. [33] implement the block-matching and 3D filtering algorithm for calculating Poisson and Gaussian noise levels. They also apply PCA for feature reduction. However, this approach typically suffers from a high false-positive rate which reduces the efficiency.

3.1.5 Pixel-based approaches

Pixel-based image detection techniques refer to detection approaches in which an image is processed per-pixel basis. This approach analyzes the properties of individual pixels, such as color, intensity, and texture, to identify objects in an image. Some standard pixel-based image detection techniques [34] are thresholding, edge detection, template matching, and Hough transform. Transformation typically maps the correlated features into uncorrelated features for the neighboring blocks. Z. Moghaddasi et al. [35] propose SVD with Discrete Cosine Transform (DCT) and Kernel PCA. Scholars also suggest Discrete Factional Cosine Transform (DFrCT) [36] in place of DCT, which evaluates the extra fractional parameters. Kernel PCA identifies the critical features as every feature is unimportant for analysis. H. Sheng et al. [37] propose Quaternion-DCT (QDCT) with a difference matrix in the quaternion domain. Quaternion Backpropagation Neural

Network (QBPNN) opts for a classifier. The DCT [38] operates on real-valued data and provides a compact representation in the frequency domain. At the same time, the QDCT uses quaternions to represent color information and captures color variations in the frequency domain.

R. Mehta et al. [39] compare three feature extraction methods: DCT, Discrete Wavelet Transform (DWT) [40], and spatial. The PCA is used for feature reduction and the ensemble classifier is used for classification, as shown in Fig. 10. X. Zhao et al. [41] present a 2D Markov model with the help of Block DCT, discrete Meyer wavelet transform, and SVM. Since the image is a 2D signal, generalizing it with a 2D Markov model makes it more accurate and robust. The basic Markov model consumes more time and hence increases computational complexity.

In 2018, S. A. Nagtode and S. A. Korde [42] resolved the angular shift and resizing of the image by Multi-Scale Local Binary Pattern (MS-LBP). MS-LBP applies on each sub-band, such as A (approximation), V (vertical), H (horizontal), and D (diagonal), to find out the features. LBP with DCT is not that effective for noise sensitivity and pixel modification.

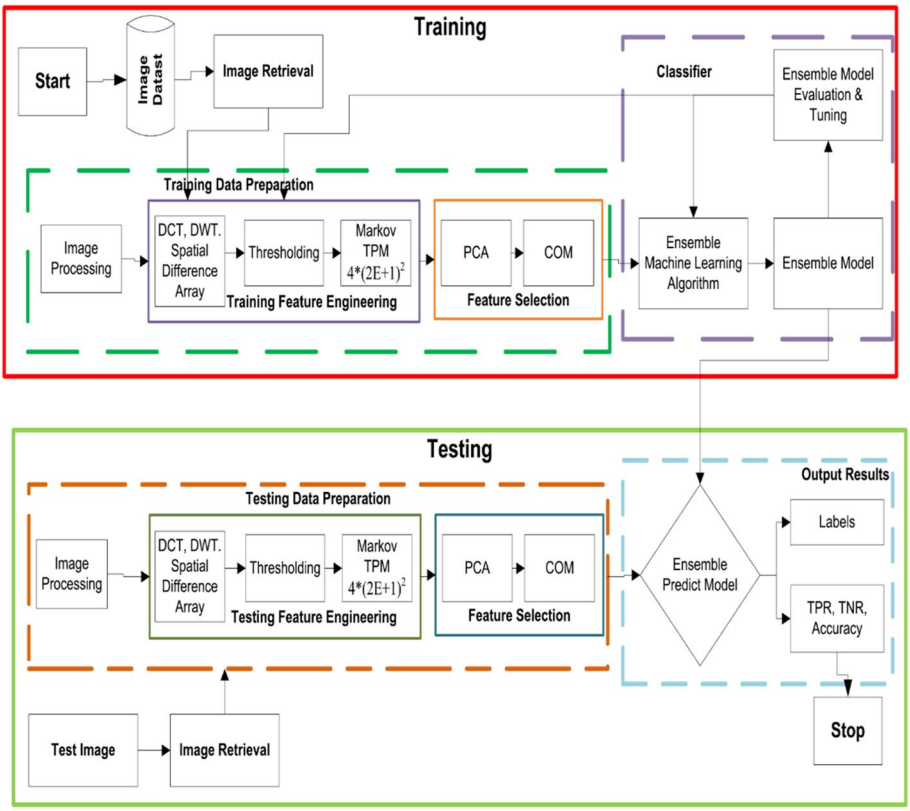

Fig. 10 Flow chart for Pixel-based splicing detection proposed by R. Mehta et al. [39]

3.1.6 Geometric-based approaches

Geometric-based detection techniques rely on the geometry or shape of an object. Whenever two or more images merge, there are obvious chances of change in geometric features such as shape, size, symmetry, and orientation of objects within an image. The basic steps to the geometry transformation are (a) extraction of ROI, (b) warping as per mapping function, (c) interpolation, (d) filtering and smoothing (e) performing the sampling.

Some standard geometric-based splicing detection techniques are Scale-Invariant Feature Transform (SIFT), Speeded Up Robust Features (SURF), Features from Accelerated Segment Test (FAST), and Random Sample Consensus (RANSAC). SIFT is a feature detection and extraction technique based on an object's scale-invariant properties. It uses a scale-space representation to identify key points in an image invariant to scale and orientation. RANSAC is a robust estimation technique that fits a geometric model to a set of data points. It is used for object detection by fitting a geometric model, such as a line or a circle, to the features of an object in an image. C. Chen et al. [43] present multiple combinations of rotation and resizing to detect splicing with the help of successive geometric transformations. E. Odabaş Yıldırım and G. Ulutaş [44] use a hybrid approach using the Stationary Wavelet Transform (SWT) and Markov model. Unlike traditional wavelet transforms based on dyadic (binary) scaling and translation, the SWT allows for flexible scale and translation parameters.

In 2018, H. Chen et al. [45] propose SLIC and FAST. SLIC is a simple linear iterative clustering algorithm that creates super-pixels for image blocking. The size of the super-pixel depends on the color variation area in any image. FAST is a popular feature detection technique to identify an image's corners. It is known for its efficiency and speed in detecting corners, which makes it suitable for real-time applications. K. B. Meena and V. Tyagi [46] suggest an adaptive Haar wavelet transform, also known as Tetrolet transforms, in place of DWT to extract geometric features. It improves accuracy by correctly differentiating the forged region and naturally identical regions.

3.1.7 Texture-based approaches

Texture in an image refers to the visual quality and appearance of the surface of objects in the image. It defines an image's repetitive and recognizable pattern of intensities or colors. It is an essential feature of an image, as it provides information about the surface properties of objects in the scene. The textural properties of the manipulated region may differ from those of the surrounding regions, which identifies the forgery. The relevant textural features are extracted using histogram, LBP [47, 48] or GLCM.

K. H. Rhee [49] extracts the textural features using median filter residual. The ground truth mask localizes the manipulation by calculating grey-level maxima and Entropy-based Edge (EbE). Lastly, cubic SVM is used for the classification of spliced areas. Jalab H. A. et al. [50] present a novel method called an approximated Machado fractional entropy (AMFE) of the DWT. The AMFE serves as a new fractional texture descriptor, while the DWT decomposes the input image into several sub-images, each representing different frequency bands. In contrast, Z. Moghaddasi et al. [51] analyze textural features through the Run Length Run Number (RLRN) algorithm. The kernel PCA reduces the dimensions and removes redundant feature information to avoid complexity, and finally, the Library for SVM (LIBSVM) classifies through a grid search method.

A. K. Jaiswal and R. Srivastava [52] combine textural and shape features. They designed a logistic regression-based ML tool to identify whether the image is spliced. However, the model is not suitable for down-sampled images. In this work, the RGB image transforms into a grayscale image then four feature extraction tools, i.e., Histogram of Oriented Gradients (HOG), Local Ternary Patterns (LTP), DWT, and LBP [53], apply to the image. DWT provides a multi-resolution representation of a signal or image, allowing for efficient compression and analysis. In contrast, DCT [54] doesn't achieve high accuracy, as it avoids the correlation between pixels. So, DWT prefers DCT for high precision and accuracy.

3.1.8 Moment and image quality metrics (IQM) based approaches

IQM evaluates an image's quality in terms of its fidelity to the original scene. These metrics typically compare the original image to a processed image, intending to quantify how much the processed image has been degraded or distorted. On the other hand, moments are mathematical functions describing an image's intensity distribution. The image is divided into several regions. If the regions are significantly less in number, then splicing accuracy degrades. Whereas if the regions are more, it increases the computational time. For an actual image, moment-based features are smooth and regular, but forgery disturbs the image's smoothness.

Z. Zhang et al. [55] compute IQM for feature extraction and implement multi-size block DCT to calculate the histogram for each sub-band. The proposed model consists of 240-D moment features. T. J. Jayan and P. Sajith Sethu [56] present a detection approach based on illuminating maps and IQM with the Naïve Bayes classifier.

3.1.9 Light-based approaches

Background and lighting properties are different in every image. In splicing, when two or more images are merged to create a new forged image, the semantic illumination or reflectance of the image breaks out. This inconsistency is used in the evaluation of forgery detection. Illuminance refers to the amount and direction of light in the scene, which is usually associated with low frequency (inside the image). In contrast, reflectance refers to the surface properties of the objects in the scene, which deals with high frequency (at cropped corners). The study [57] presents an approach with SVM and the Viola-Jones algorithm. This algorithm is used due to its high detection accuracy, robustness against lighting and facial expression variations, and computational efficiency.

P. Niyishaka and C. Bhagvati [58] introduce an illumination-reflectance model to calculate illuminance and chrominance components. The LBP is used for feature extraction, and multiple classifiers SVM, Linear Discriminant Analysis (LDA), logistic regression, K- Nearest Neighbor (KNN), Decision Tree, and Naïve Bayes are used. F. Hakimi et al. [59] developed a scheme using DWT and SVM. However, to improve results, the authors in their other publication [60] replaced DWT with DCT for feature extraction and SVM with KNN as a classifier. The study proposes a passive splicing detection scheme using Improved LBP and DCT. It divides the chrominance component into non-overlapping blocks, calculates Improved LBP, and transforms it using 2D-DCT. Later, the KNN is employed for classification.

Most of the other studies, such as [18, 32, 48, 55, 61], show that SVM is the most preferred technique by scholars for binary or two-class classification in image splicing. Naïve Bayes classifies the data by a set of probabilities, where the probability of one feature

is completely ineffective from another. On the other hand, KNN calculates the distance between feature vectors. However, its data computation is time-consuming and costly.

Most of the abovementioned techniques are based on 2D images but in a very different approach. B. Peng et al. [62] suggest extracting the 3D pose of a standing object using a 3D morphable model. Table 2 compares all hand-crafted methods that work on the Chinese Academy of Sciences Institute of Automation (CASIA), and Table 3 tabulates COLUMBIA and Table 4 lists other customized datasets to compare the performance of models based on accuracy. The table's bold reference highlights the model with the highest accuracy. The localization of splicing in an image is also listed as image level, block level, pixel level, and object level.

3.2 Deep learning-based splicing detection approaches

This section incorporates detection approaches based on deep learning, as they have exhibited promising outcomes in effectively pinpointing and localizing image-splicing manipulations. These approaches are popular among scholars due to their high accuracy and ability to learn complex features from large datasets. CNNs [63–66] are a popular deep-learning approach for image analysis tasks, including image splicing detection. The approach involves training a neural network on a dataset of spliced and authentic images to learn the features that differentiate them. Once trained, the CNN can classify new images as spliced or authentic. It performs both feature extraction and classification simultaneously. It uses a series of convolutional layers to extract features from input data, and then these features are fed into fully connected layers that perform classification. Examples of CNN architectures are GoogLeNet, MobileNet [67], DenseNet [68], and VGA16 [69]. The basic CNN architecture is shown in Fig. 11, where preprocessing, convolution, and pooling are the necessary steps.

D. Kim and H.Y. Lee [63] present a CNN model augmented with a high pass filter to extract hidden features. This model contains local response normalized layers, two max-pooling and fully connected layers, and a high-pass filter. The authors [70] propose the combination of CNN, PCA, and DWT to detect splicing. They categorize image splicing into three main parts: LBP, Markov, and Deep Learning. Abhishek and N. Jindal [71] apply deep CNN and semantic segmentation, and Y. Rao et al. [72] use CNN-128 as a local descriptor and SVM classifier for localization. When referring to "CNN-128," it signifies that the final layer of the convolutional neural network has a dimensionality of 128. This dimensionality directly affects the richness and complexity of the learned features, thereby playing a crucial role in determining the network's performance.

L. Almawas et al. [73] compare the results of SVM, KNN, and Naïve Bayes, where KNN performs the best and achieves higher accuracy and sensitivity rate. Furthermore, CNN extracts the features. At the same time, the authors in [74] combine the weighted sum for three features, YC_bC_r, Photo Response Non-Uniformity (PNRU), and edge features, to train the CNN model for achieving high accuracy and less redundant information. PRNU explains noise, YC_bC_r characterizes the color, and the Edge feature [75] works with pixel inconsistencies. B. Chen et al. [76] propose the model with three FCN (fully convolutional networks), FCN8, FCN16, and FCN 32, and the (Long Short-Term Memory) LSTM with Region Proposal Network (RPN) for sampling features. By removing the fully connected layers, FCNs preserve the spatial information of the input image throughout the network, enabling pixel-level predictions. LSTM is also proposed in [77].

Table 2 The accuracy outcomes of handcrafted techniques on the CASIA dataset

S. No	Reference	Year	Localization	Classifiers	Feature Extraction Technique	Accuracy
1.	Z. Moghaddasi et al. [51]	2014	Image-level	LIBSVM	RLRN and PCA	88.28% 90.38%
2.	Z. Moghaddasi et al. [35]	2018	Block level	SVD	2D-DCT	99.30% 97.15% 96.97%
3.	H. Sheng et al. [37]	2018	Block level	Q-BPNN	Q-DIFF and Q-Markov4D	99.21% 98.66%
4.	T. Subramaniam et al. [61]	2019	Image-level	SVM	RDWT	98.30%
5.	Jalab H. A. et al. [50]	2019	Image-level	SVM	DWT, Fractional entropy	99.50%
6.	**A. K. Jaiswal and R. Srivastava** [52]	**2020**	**Image-level**	**Logistic Reg**	**HoG** **LTP** **DWT** **LBP**	**98.6%** **98.3%** **99.5%**
7.	P. Niyishaka and C. Bhagvati [58]	2021	Image-level	SVM LDA Logistic Reg KNN Decision Tree Naïve Bayes	LBP Illumination and Chrominance	91.45% 93.79% 90.82% 86.08% 91.07% 76.93%

Table 3 The accuracy outcomes of handcrafted techniques on the COLUMBIA (DVMM) dataset

S. No	Reference	Year	Localization	Classifiers	Feature Extraction	Accuracy
1.	Z. Zhang et al. [55]	2008	Image-level	SVM	MBDCT, IQM	87.10%
2.	H. Chen et al. [45]	2018	Pixel-level	–	SLIC + FAST	–
3.	X. Zhao et al. [41]	2015	Block-level	SVM	BDCT + Meyer WT	93.36%
4.	**R. Mehta et al.** [39]	**2021**	**Pixel-level**	**Ensemble Classifier**	**DCT, DWT, Spatial PCA**	**99.99%**

Table 4 The accuracy outcomes of handcrafted techniques on the customized dataset

S. No	Reference	Year	Localization	Classifiers	Feature Extraction	Accuracy
1.	B. Yang et al. [18]	2015	Object-level	SVM	GRPW	88.54% (TPR)
2.	K. Bahrami et al. [19]	2015	Pixel-level	SVM	MAP	95.3%
3.	B. Liu and C. M. Pun [32]	2020	Pixel-level	SVM	ASVD	99.82%
4.	**K. H. Rhee** [49]	**2020**	**Block-level**	**Cubic SVM**	**Gray-level regional Maxima and EbE**	**99.99%**
5.	T. J. Jayan and P. Sajith Sethu [56]	2018	Image-level	Naïve Bayes	IQM, MAP	95.00%

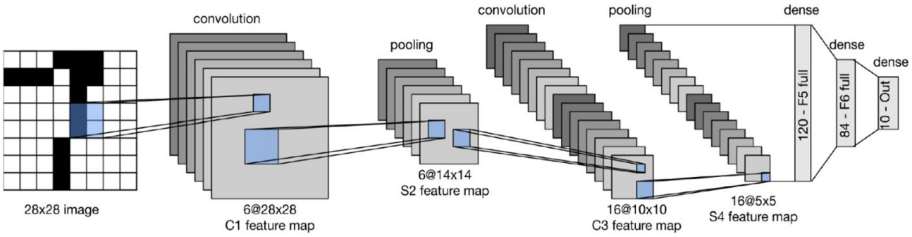

Fig. 11 A basic CNN architecture proposed by J. M. Czum et al. [61]

Other deep learning architectures, such as Circular U-Network [78], Deep Coupled U-Network (DCU Net) [79], Coarse-to-Refined CNN (C2RNet) [80], and Residual Neural Networks (ResNet)-50 [81] are also popular in splicing detection. H. Ding et al. [79] apply DCU Net to locate the manipulation through pixel-based detection. The features obtained from the dual-channel encoding network are combined, and subsequently, tampered features with varying levels of detail are extracted using dilation convolution. Finally, a secondary fusion is performed. B. Xiao et al. [74] propose a C2RN, a progressive deep NN with a diluted adaptive clustering. The C2RNet progressively refines the output by starting with a coarse representation and gradually enhancing it through subsequent layers to extract finer details. In parallel, diluted adaptive clustering is employed to adaptively group similar data points based on their features, enabling more efficient and effective data analysis.

The study [82] proposes Mask Region-based CNN (Mask-RCNN), a deep learning model designed for object detection and instance segmentation in images. It combines

object detection and semantic segmentation tasks into a two-stage process. In a similar approach, S. Nath and R. Naskar [81] combine CNN with an ANN classifier. They also apply ResNet-50 to extract features from the input image. However, the model doesn't precisely locate the manipulated area. X. Bi et al. [83] introduce a novel approach called the ringed residual U-Net (RRU-Net) for enhancing the learning process of CNN. The RRU-Net enables a ring structure formed by connecting feature maps from different scales and stages in a circular manner.

In the context of detecting fake and deepfake images, one effective approach is Generative Adversarial Networks (GANs) [84, 85]. A basic architecture of GAN is presented in Fig. 12. These networks are made of two artificial neural networks: one is the generative network and the other is the discriminative network. With the help of a real dataset, the generative network trains itself and produces images that are very similar to the original dataset. On the other hand, the discriminator network is trained to distinguish between real dataset images and the images produced by the generative network.

The other types of GANs are CycleGAN [86], Deep Convolutional GAN (DCGAN) [87], Least Squares GAN (LSGAN) [88], BigGAN [89], and Progressive Growing of GAN [90]. GANs and CNNs both are deep learning networks but they differ in their architecture. A GAN can be made with two CNNs. GANs are designed to create new fake images from the existing dataset whereas the CNNs are mainly employed for classification and recognition.

Based on the GANs, Kniaz V. et al. [91] present two generative models (retouching and annotation) and two discriminator models (again retouching and annotation). The retouching models are responsible for translating manipulated images into the real image domain and the annotation model estimates the pixel-wise probability of image patches, determining whether the images belong to the real or spliced category. The methodology involves training the generative retouching model against a deep semantic segmentation generative annotation network.

The authors in [92] specifically introduce Reality Transform GANs with transformation and localization generators. The transformation generator incorporates an alpha-learnable Whitening and Coloring Transformation (WCT). Through adversarial training, this block automatically suppresses tampering artifacts in forged images. The

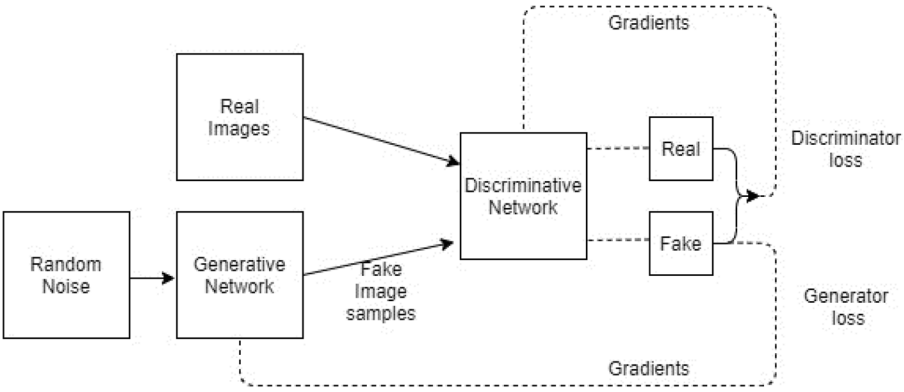

Fig. 12 Basic architecture of GANs

localization generator adopts a multi-decoder-single-task strategy to identify the tampered regions within the images accurately.

Constrained Image Splicing Detection and Localization (CISDL) [93] is based on the idea that splicing introduces specific statistical changes into the forged image that are not present in the original images. For splicing localization, Y. Liu et al. [93] presented a GAN-based Deep Matching network using Atrous Convolution (DMAC). The hierarchical features extracted from the artous convolution are fed to a detective and discriminative network. The data from these networks is then used to train the DMAC to find out the matching in the two images. The model is limited to fixed-size images only. Therefore, to localize pixel-level splicing on high-resolution images, the authors [94] extended their work by introducing a novel approach called Attention-aware encoder-decoder Deep Matching network (Attention DM). The two networks VGG and ResNets are used for feature extraction.

Scholars have recently proposed various approaches to deep learning-based image splicing detection. The existing approaches are presented in Table 5 and 6. The standard datasets **CASIA v1.0 & v 2.0** and **DVMM** are chosen to compare performance in terms of accuracy. The approach with the highest accuracy is highlighted in bold.

4 Dataset

In this section, several publicly available datasets for image splicing detection and localization are discussed in detail. However, it is important to note that each dataset has its strengths and weaknesses, and researchers should choose the dataset that best fits their research needs.

Some datasets, such as CASIA 1.0, DSO-1, DSI-1, Columbia, etc., are available online. In these datasets, the spliced images are generated by cutting and pasting, while some are retouched and post-processed with various attacks. However, most datasets focus on one or two forgeries and support fewer image formats. So, there is a need for a generalized dataset that contains all attacks and supports multiple image formats like PNG, JPEG, TIFF, WEBP (Image file format developed by Google), and DNG. Some of the datasets specially designed for splicing detection are given below.

COLUMBIA is known as the Columbia uncompressed image splicing detection evaluation dataset [95] provided by DVMM Laboratory and is open to all. The data is available in two versions; one in color (CUISDE) with 757×568 to 1152×768 pixels size, and the other in grayscale with 128×128 pixels size.

CASIA v1.0 is made by the Chinese Academy of Sciences' Institute of Automation [96]. It has 800 original and 921 manipulated images in JPEG format only. The pixel size is 374×256, and all the images are grayscale.

CASIA v 2.0 consists of 12,323 colored images with different Q factors. This dataset [97] uses both copy-move and splicing. It has 7491 authentic images and 5123 forged images. It is also known as TIDEv2.0.

NIST NC 2016 contains forgeries like copy-move, splicing, and removal. It was introduced in 2016 and is not available for all. It has 564 forged and 560 authentic images in total [98].

Media Forensic Challenge (MFC) 2018 [99, 100] dataset contains 35 million images and 300,000 small video clips for different forgeries. It includes forensics, image processing, imaging, computer vision, etc. It contains crop, slicing, cloning, shadow, reflection, and other inconsistencies.

Table 5 The accuracy outcomes of deep learning techniques on the CASIA dataset

S. No	Reference	Year	Localization	Classifiers	Feature Extraction Technique	Accuracy
1.	B. Chen et al. [76]	2018	Pixel-based	RPN	FCN8, FCN16 and FCN32, LSTM	–
2.	Y.Liu et al. [94]	2019	Pixel-based	GAN	Atrous convolution	92.55%
3.	Y.Liu et al. [93]	2020	Pixel-based	Attention DM	VGG, ResNet-50, 100	96.76%
4.	B. Xiao et al. [80]	2019	Pixel-based	C2RNet	Adaptive filtering	–
5.	Kniaz V. et al. [91]	2019	Pixel-based	GAN	Deep Semantic Segmentation	74.01% (P)
6.	E. I. Abd El-Latif et al. [70]	2020	Image-based	SVM	CNN-DWT	97.27%
7.	Y. Rao et al. [72]	2020	Block-based	SVM	CNN-128	97.50%
8.	L. Almawas et al. [73]	2020	Image-based	CNN	SVM, KNN, Naïve Bayes	86.5% 88.6% 87.01%
9.	**J. Wang et al.** [74]	**2020**	Pixel-based	**CNN**	**YC_bC_rEdge Detection, PRNU**	**99.45%** **99.32%**
10.	H. Ding et al. [79]	2021	Pixel-based	CNN	DCU Net	97.93%
11.	S. Nath and R. Naskar [81]	2021	Image-based	SVM	ResNet-50, CNN + ANN	96.45%
12.	X. Bi et al. [92]	2021	Block-based	GAN	WCT	70.70% (P)

Table 6 The accuracy outcomes of deep learning techniques on the DVMM dataset

S. No	Reference	Year	Localization	Classifiers	Feature Extraction Technique	Accuracy
1.	**Abhishek and N. Jindal [71]**	**2020**	**Pixel-based**	**DCNN**	**Semantic Segmentation**	**98.69%**
2.	Y. Rao et al. [72]	2020	Block-based	SVM	CNN-128	97.50%

Synthetic is designed for splicing but is not publicly available. It contains 65,000 images in total with 1024×1024 image size. [77]

Common Objects in Context (COCO), a large-scale dataset [101, 102] contains 328,000 images, out of which 200,000 are forged under various forgeries. It is available for all on Microsoft.

PS Battles [103] introduced in 2018, is a dataset that comprises a diverse collection of multiple attacks, consisting of 102,028 forged images and 11,142 authentic images. The dataset includes images in both PNG and JPEG formats, with varying pixel sizes ranging from 130×60 to 10,000×8558.

BSDS300 consists of 300 color and grayscale images in BMP format. The resolution is 481×321 pixels. [104]

DSO-1 and DSI-1 [79] are composite datasets of facial images. DSO-1 comprises 200 images with a resolution of 2048×1536. Among these, 100 images are authentic, while the other 100 are forged. On the other hand, DSI-1 includes 50 images, with 25 being genuine and 25 being manipulated. Additionally, this dataset incorporates retouching techniques.

LibRaw, an open library published in 2008, contains raw files and images. The images support various forgeries like geometry, CFA, Bayer pattern, white balance, etc.[105]

Image Manipulation Dataset (IMD) contains 160 forged images and 48 base images and is available on their website. It contains rotation, scaling, copy-move, and other attacks. It contains the images with JPEG compression and resampling. [106]

Multiple Images Splicing Dataset (MISD) [94] designs for especially splicing detection. It contains the images in JPG format, where forged images are 300 and original images are 618.

Dresden consists of 13,000 images captured from more than 18 camera models. All images are in JPEG format and comply with various forgeries [107].

Above mentioned datasets are also tabulated in Table 7 with additional details like the total number of real/ forged images, supporting image formats, and types of forgeries.

5 Solution approaches for image splicing forgery detection

In this section, the most preferred solution approaches and their limitations are discussed. The classification of solution approaches based on feature analysis is categorized into three main categories: block-based, key feature-based, and learning-based approaches. These categories are depicted in Fig. 13.

Table 7 List of datasets available for image splicing forgery detection

S. No	Dataset	Type of Forgery	Forged Images	Real Images`	Resolution	Image Format	Limitation
1.	Columbia Gray (DVMM v1) [95]	Splicing	912	933	128 × 128	BMP	This dataset excludes compressed images taken by the camera during image acquisition
2.	Columbia Color (CUISDE) [95]	Splicing	180	182	757 × 568 1152 × 768	TIFF, BMP	This dataset does not include the images with rotational attacks
3.	CASIA v1.0 [96]	Splicing + CM	921	800	374 × 256	JPEG	Only contain RGB images
4.	CASIA v2.0 [97]	Splicing + CM	5123	7491	320 × 240 to 800 × 600	JPEG, BMP, TIF	Ground truth masks are not given
5.	NIST NC 2016 [98]	Splicing + CM	564	560	500 × 500 5616 × 3744	JPEG	Contains high-resolution images which require heavy GPUs
6.	Synthetic [77]	Splicing + CM	4800	–	–	PNG, TIF, BMP, JPEG	Synthesized Dataset containing images with ground-truth binary mask only
7.	DSO-1 [108]	Splicing	100	100	2048 × 1536	TIFF	Images do not support JPEG format
8.	DSI-1 [109]	Splicing	25	25	–	PNG	Very small dataset which may not be able show the accuracy of the model
9.	BSDS300 [104]	Various	300		481 × 321	BMP	–
10.	IMD [106]	Splicing + CM	160	48	–	JPEG	The quantization table is not diverse
11.	COCO [102]	Various	200,000	330,000	–	–	Do not contain post-processed images
12.	PS Battles [103]	Various	102,028	11,142	130 × 60 to 10,000 × 8558	PNG, JPEG	–
13.	MISD [67]	Splicing	300	618	384 × 256	JPG	The dataset do not contain post-processed images

Table 7 (continued)

S. No	Dataset	Type of Forgery	Forged Images	Real Images`	Resolution	Image Format	Limitation
14.	LibRaw [105]	Splicing	400	20	–	DNG, RAW	The forged area is very small (1/16 of the real image) therefore it is hard to localize the forging
15.	Dresden [107]	Various	13,000	–	4000×3000	JPEG	Images in this dataset are captured with 53 cameras from 18 different models. It is a bit challenging to classify forged images captured with different camera models

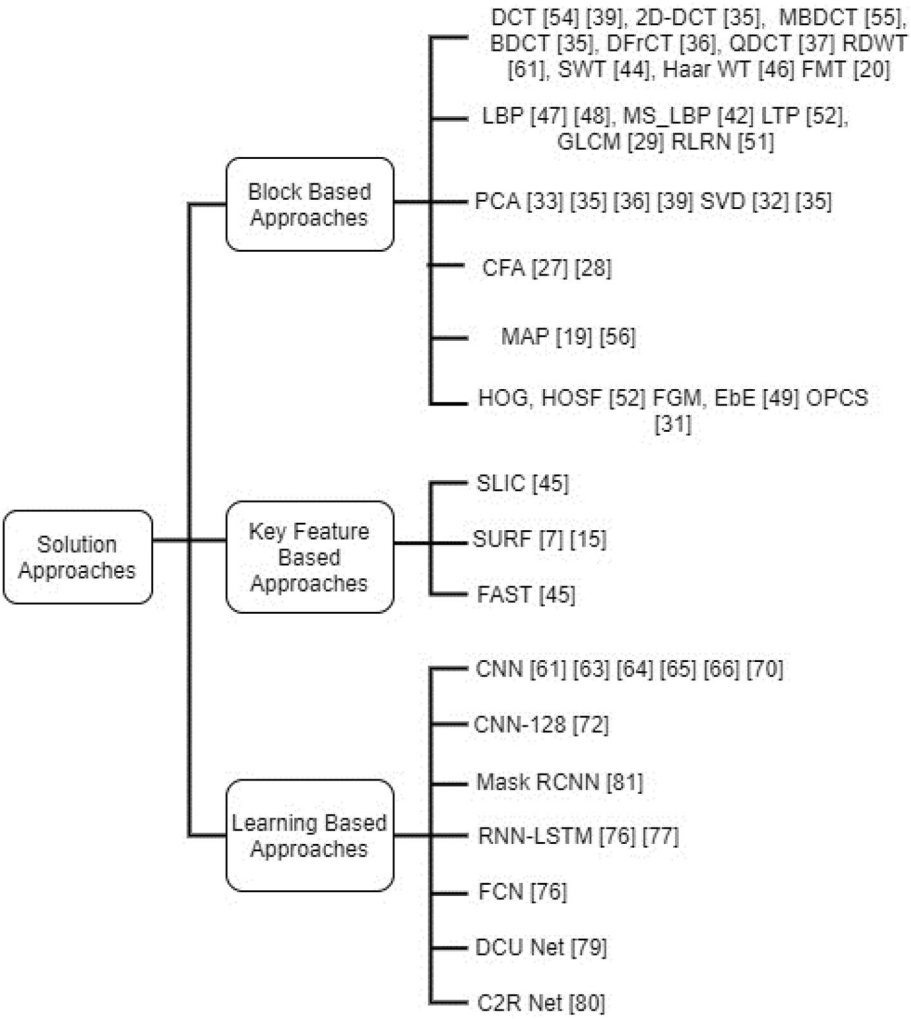

Fig. 13 Classification of various solution approaches based on feature analysis

5.1 Block-based approaches

Block-based approaches involve dividing the image or data sequence into smaller blocks or regions and analyzing the features within those blocks. Various approaches are used to extract feature information listed below;

5.1.1 DCT

It transforms the image or data sequence from the spatial domain to the frequency domain, emphasizing different frequency components. These DCT coefficients capture the

frequency information in an image, and the changes in these coefficients can indicate the presence of forgeries. However, the patch-wise merging of information in a DCT histogram results in a loss of its visual representation.

5.1.2 DWT & SWT

The DWT and SWT are popular multi-resolution transforms that decompose images into different frequency sub-bands. These transforms help detect different types of image forgeries. The high-frequency sub-bands reveal artifacts or inconsistencies introduced by image manipulations, while the low-frequency sub-bands provide information about the overall structure and smoothness of the image. SWT may reduce some of the limitations of DWT.

5.1.3 FMT

It combines the Fourier Transform and the Mellin Transform to analyze and extract rotational and scale-invariant features from images. It is known for its ability to handle scale and rotation variations and provide robust features for image analysis. However, it may have limitations in other deformations or non-linear transformations. Various other transforms have been explored for splicing detection, including the Fractal-Markov Transform, Curvelet Transform, and Radon Transform. These transforms have specific properties that make them suitable for certain forgeries.

5.1.4 LBP

It is a simple yet powerful texture descriptor that characterizes an image's local patterns of pixel values. It computes a binary code for each pixel by comparing its intensity value with neighboring pixels. The resulting binary patterns are then used to construct histograms that capture the frequency of patterns occurring within the image. It is important to note that LBP captures local texture information by comparing the central pixel with its neighboring pixels. It is robust to image transformations such as rotation and scaling. It constructs histograms to represent the distribution of different local patterns.

5.1.5 LTP

It captures more detailed texture variations than LBP by considering three intensity levels and generating ternary codes. LTP allows for a richer representation of local texture patterns in an image.

5.1.6 GLCM

A statistical method that captures the spatial relationship between pairs of pixels based on their grey-level values. It calculates the occurrence of different combinations of grey-level values in an image's specified spatial relationship (such as distance and angle). It considers the entire image and quantifies the relationship between pixel pairs. From the GLCM matrix, various statistical measures like contrast, energy, correlation, and entropy can be derived to describe the texture properties.

5.1.7 RLRN

An algorithm that analyzes textural features by considering the runs of consecutive elements with the same value in an image or data sequence. It focuses on extracting statistical information about the spatial distribution of pixel values within the image or sequence. It computes statistical measures such as mean, variance, entropy, and energy from the RLRN to represent the textural properties.

5.1.8 PCA

A linear transformation technique aims to find the orthogonal axes, known as principal components, along which the data exhibits the maximum variance. It is primarily used for analyzing and reducing the dimensionality of data. It finds a new set of principal components that capture the most critical information in the data. The principal components are ranked in order of decreasing variance, allowing for dimensionality reduction by selecting a subset of the components.

5.1.9 SVD

It is a matrix factorization technique that decomposes a matrix into three components: an orthogonal matrix, a diagonal matrix with singular values on the diagonal, and the transpose of the orthogonal matrix. SVD decomposes a matrix into its constituent parts, providing insights into the matrix's rank, singular values, and singular vectors.

5.1.10 CFA

It is a mosaic pattern of color filters used in digital image sensors to capture color information. The most common CFA is the Bayer filter, which arranges red, green, and blue filters in a repeating pattern over the pixels of an image sensor. CFA allows digital cameras to capture color information despite having image sensors that inherently only detect grayscale intensity. The purpose of a CFA is to enable the image sensor, which inherently detects grayscale intensity, to capture color information.

5.2 Key-feature based approaches

Key-feature approaches aim to identify and extract specific features or key points within the image or data sequence representing important textural information. Two commonly used techniques are:

5.2.1 SIFT

It detects key points by identifying scale-invariant and rotation-invariant features in an image. It uses a Difference of Gaussian (DoG) pyramid to detect key points at different scales and orientations. SIFT descriptors are computed based on the local image

gradients around the key points based on the presence of stable features, such as corners, blobs, or edges.

5.2.2 SURF

It detects key points at multiple scales and uses the Hessian matrix approximation technique for efficient detection. SURF descriptors are based on the sum of Haar wavelet responses within a local neighborhood around the key points regions with high principal curvature.

5.2.3 FAST

It is designed to be a faster algorithm than SIFT and SURF. It achieves computational efficiency using simple intensity comparisons and a corner detection mechanism. But SIFT and SURF provide more informative and distinctive feature descriptors than FAST. SIFT and SURF are known for their robustness to scale changes and rotation, while FAST is primarily designed for detecting corners and is less invariant to scale and rotation. However, a better alternative algorithm called ORB (Oriented FAST and Rotated BRIEF) is available. ORB combines FAST corner detection with BRIEF descriptors and has gained popularity as a viable alternative to SIFT and SURF, providing a good balance between speed and accuracy.

The feature maps of several hand-crafted techniques are combined in Fig. 14 to give an idea about the selection principle behind these approaches.

5.3 Learning-based approaches

Learning-based approaches leverage the power of deep learning algorithms to learn features directly from the data automatically. Some commonly used deep learning techniques include:

5.3.1 CNN

CNNs are widely used in splicing detection because they excel at extracting spatial features from images effectively. This characteristic is vital for identifying forged regions within an image. By utilizing convolutional layers for feature extraction and pooling layers for downsampling, CNNs can capture local patterns and create hierarchical representations, enabling them to detect manipulations at various levels of abstraction. One key principle for selecting CNNs in splicing detection is their ability to handle spatial information adeptly. When comparing their performance, CNNs tend to shine in scenarios where the forgery detection task relies heavily on spatial features. However, their effectiveness may vary depending on the complexity of the manipulations and the size and diversity of the dataset. In cases where local patterns and spatial characteristics are the primary cues for forgery detection, CNNs often outperform other classifiers.

5.3.2 RNN

RNNs, particularly variants like Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU), are specifically designed for sequential data processing, making them a

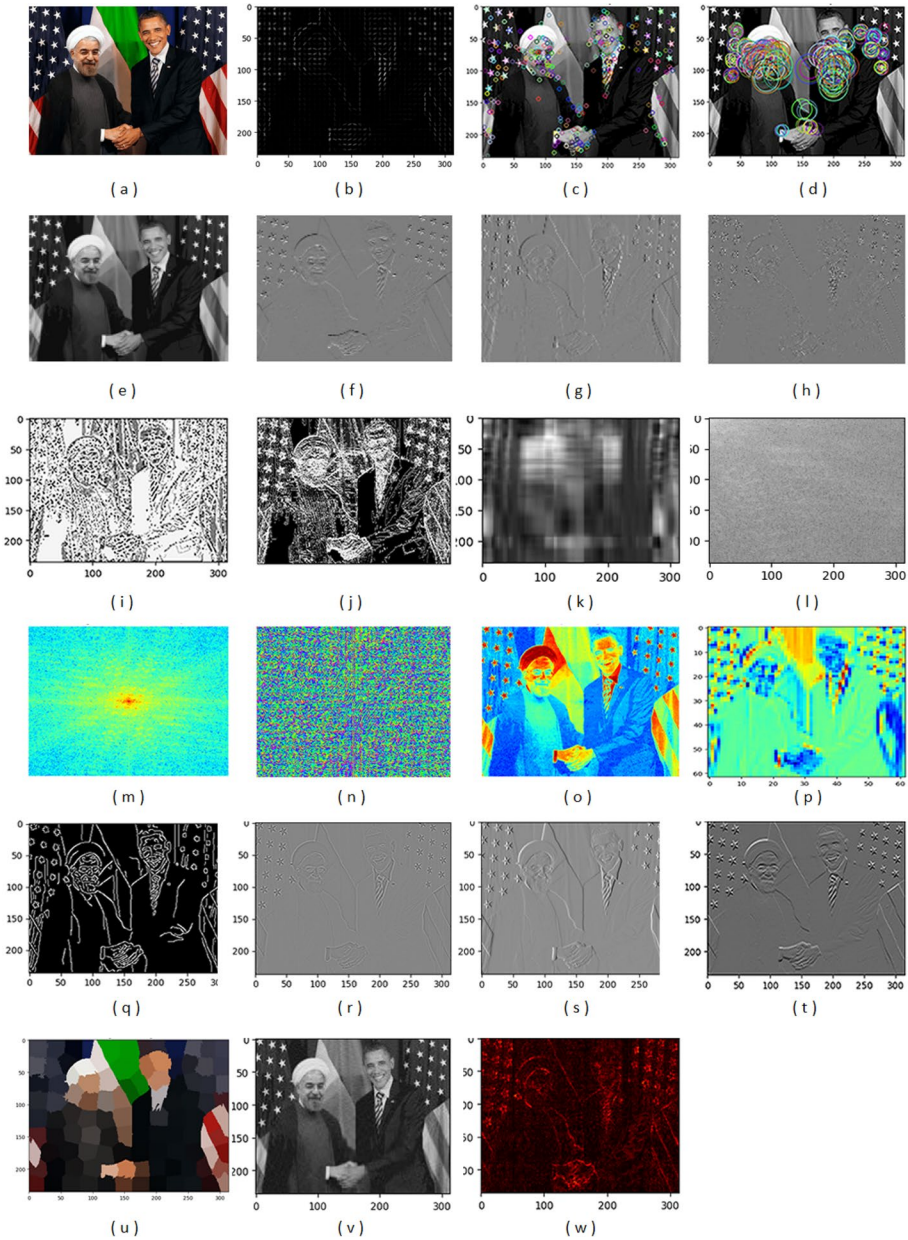

Fig. 14 Example of feature extractions from several hand-crafted techniques (**a**) Original image, (**b**) HOG features map, (**c**) SIFT feature map, (**d**) SURF feature map, (**e**) DWT-LL Pattern, (**f**) DWT-LH pattern, (**g**) DWT-HL pattern, (**h**) DWT-HH pattern, (**i**) LBP (**j**) LTP (k) PCA (**l**) DCT (**m**) SMFT- Magnitude feature map (**n**) SMFT-Phase feature map, (**o**) SMFT-Overlayed magnitude (**p**) Q-Diff feature map, (**q**) IQM-Laplacian edge map (**r**) IQM-sharpness map (**s**) IQM-contrast map (**t**) IQM-brightness map (**u**) SLIC (**v**) SVD-reconstructed image, (**w**) SVD-feature map

valuable choice for splicing detection tasks involving temporal dependencies or time series analysis. When the forgery detection problem necessitates the consideration of temporal sequences or dynamic changes within an image, RNNs become a compelling option. However, it's essential to note that their selection should be contingent upon the presence of significant sequential patterns. In scenarios where spatial information plays a more crucial role than temporal dependencies, alternative classifiers may offer better performance. Therefore, when comparing their effectiveness, RNNs demonstrate their strengths in tasks where temporal patterns are prominent but may lag behind in cases where spatial context dominates the detection process.

5.3.3 FCN

For tasks involving splicing detection, particularly those that demand precise localization of manipulated regions within an image, Fully Convolutional Networks (FCNs) stand out as a strong choice. Their primary selection principle lies in their design for semantic segmentation, where every pixel in an image is accurately labeled. Furthermore, FCNs possess the advantage of being able to accommodate input images of arbitrary sizes, making them adaptable to scenarios with varying image resolutions. In performance comparisons, FCNs consistently demonstrate their effectiveness when the goal is to precisely identify forged regions within an image. Their pixel-level segmentation capability positions them as a top contender for splicing detection, especially in applications where granular localization is paramount.

5.3.4 R2C Net

R2C Net is a specialized architecture tailored for the detection of rotation-based forgeries, making it a standout choice in scenarios where this form of manipulation is common. It incorporates robust modules designed explicitly for identifying forged regions, even in instances where the image has undergone rotation. By combining Convolutional Neural Networks (CNNs) and attention mechanisms, R2C Net can effectively extract features and address the challenges posed by rotational variations in input data. Performance-wise, R2C Net excels in situations where image rotations are prevalent manipulation techniques. Its specialized architecture and attention mechanisms equip it with the necessary tools to identify and pinpoint forgeries that involve rotation. However, in scenarios where rotational manipulation is not a dominant factor, other classifiers might offer better overall performance.

5.3.5 DCU Net

DCU Net operates on complex-valued data, capturing both magnitude and phase information, which proves invaluable for certain types of forgeries where both amplitude and phase contain critical information. It combines this unique capability with the U-Net architecture, which excels at precise localization of forged regions. This dual approach makes DCU Net a compelling choice for segmentation-based splicing detection tasks. When assessing its performance, DCU Net shines in scenarios where the manipulation involves complex-valued information, such as certain types of image alterations that affect both magnitude and phase. In such cases, its combination of complex-valued data processing and U-Net architecture makes it a formidable contender. However, its performance may not be as robust

when dealing with simpler manipulations that do not require the handling of complex-valued data, highlighting the importance of selecting classifiers based on the specific characteristics of the forgery detection task at hand.

6 Conclusion

This review presents a detailed study on image splicing forgery detection over the past decade from 2014 to 2023, primarily identifying robust approaches in this field. The study is categorized into two main approaches: hand-crafted and deep-learning approaches, each distinguished by their feature selection. Hand-crafted approaches encompass conventional splicing detection practices that rely on manual feature extraction and subsequent analysis. These approaches involve a meticulous examination of specific attributes within an image to identify potential areas of tampering. Key features used for detection in images include aspects such as noise levels, lighting conditions, camera specifications, image formats, contextual scenarios, geometric patterns, textures, and individual pixel characteristics. In contrast, deep-learning techniques employ advanced neural network architectures for splicing detection. This category includes models such as GoogLeNet, MobileNet, DenseNet, VGA16, Circular U-Network, deep coupled U-Network, coarse-to-refined CNN, residual neural networks-50, and generative adversarial networks. The solution approaches based on feature extraction are also categorized into block-based, key feature-based, and learning-based approaches. The percentage distribution of existing work on splicing detection is presented in Fig. 15, where the pixel-based splicing detection approaches are extensively employed in numerous studies, while textural and scenario-based schemes are widely recognized and utilized for detecting splicing.

Machine learning approaches, particularly SVM and ANN, are highly preferred in forgery detection. Current research also emphasizes feature selection and reduction techniques.

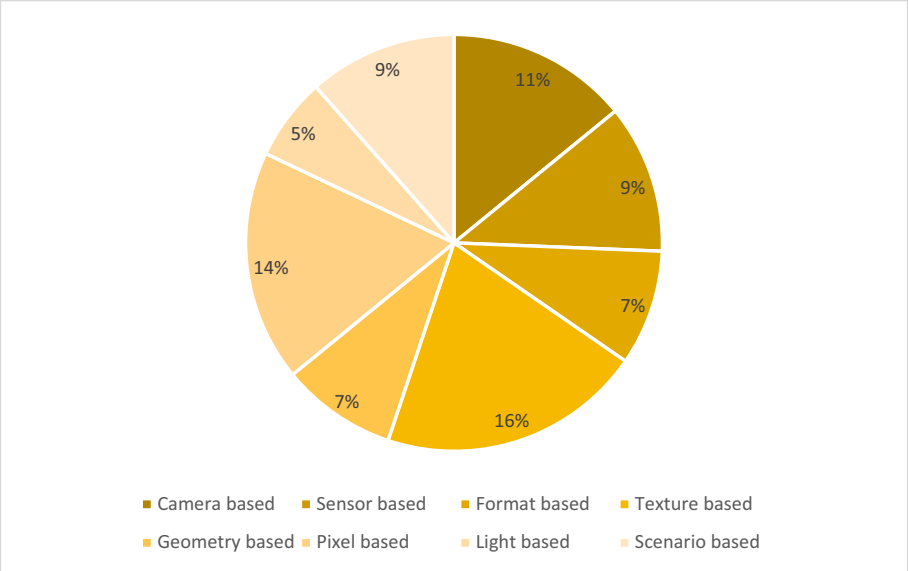

Fig. 15 Percentage-wise representation of work done on image splicing detection

The purpose of this survey is to explore the current state-of-the-art methodologies and seek insights to address the following research questions mentioned in Section 1.

RQ1: Which hand-crafted approach is the most effective for localization of splicing?

Regarding hand-crafted techniques, the approach described in reference [45] demonstrates the highest accuracy when applied to the CASIA dataset. This method relies on the combined use of Discrete Wavelet Transform (DWT) and logistic regression, proving to be exceptionally effective in pinpointing forged regions within an image.

RQ2: Which Deep-learning approach is the most effective for localization of splicing?

CNN models have demonstrated promising results in forgery detection, leveraging their inherent ability to learn automatically. These models enable easy, fast, and accurate processing. Within the domain of deep-learning networks, reference [69] introduces a notable method. It employs a CNN architecture coupled with photo response non-uniformity and edge detection-based features. Remarkably, this approach achieves an accuracy rate of 99.99% in the detection of forged areas within images. The other effective approaches are DCNN [64] with 98.48% of accuracy.

RQ3: Which classifier is the most effective for the classification of spliced images?

SVM is used the most for image-level classification between real and spliced images. But it can only handle fewer features and is a bit time-consuming. The other option is to use an ensemble classifier [rachana] that can handle high-dimensional features with higher accuracy than SVM.

RQ4: Which dataset is suitable for splicing?

This paper incorporates a variety of datasets that are pertinent to splicing detection. These datasets encompass Columbia Gray (DVMM v1), CASIA v1.0 and v2.0, DSO-1, DSI-1, COCO, BSDS300, and several additional datasets. These datasets are often curated and may not encompass the full range of potential challenges that represent real-world scenarios accurately. From the literature, it is found that Casia v2.0 is used in most research. But it contains RGB images only. Columbia Gray and DSO-1 could be an option as these datasets are specifically designed for image splicing. However, both datasets are limited to specific attacks.

RQ5: What parameter should be chosen for evaluation metrics?

True positive and true negative rates are the two parameters used for image classification to know whether the image is spliced or not. These can later be used to calculate accuracy, precision, and f-score.

7 Future work

Splicing detection has made notable advancements after introducing deep neural networks. However, there remain numerous potential avenues for researchers to pursue to enhance the accuracy and resilience of these detection approaches. Adversarial attacks can deceive deep neural networks into misclassifying manipulated images. Developing robust defense mechanisms against adversarial attacks would enhance the reliability of splicing forgery detection models. Incorporating multiple modalities, such as text or audio, alongside image data can enhance the accuracy and reliability of detecting splicing forgeries. The availability of diverse and large-scale benchmark datasets designed explicitly for splicing detection can facilitate more comprehensive evaluations of different methods. By addressing these challenges, more accurate, robust, and practical solutions may be strived in detecting spliced image manipulations.

Data Availability It is not applicable to this article as no datasets were generated or analyzed during the current study.

Declarations

Competing interests The authors declare that they have no conflict of interest.

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